Pre-Characterization Process Risk Assessment

A-Mab Drug Substance — A-Mab Drug Substance — RA-001

Process Development / Manufacturing Science & Technology (MSAT)

2026-07-24

Table of contents

Field	Value
Document ID	RA-001
Document class	Pre-Characterization Process Risk Assessment
Title	Pre-Characterization Process Risk Assessment: A-Mab Drug Substance
Product	A-Mab
Version	1.0
Effective date	2026-07-24
Status	Approved (synthetic)

Field	Value
Document owner	Manufacturing Science & Technology (MSAT)
Sponsor / sites	Novacyte Biologics; Novacyte Biologics — Cambridge, MA (Development) → Novacyte Biologics — Grafton, WI (Commercial DS)

SYNTHETIC DOCUMENT — NOT A REAL RECORD. Illustrative content generated from a seeded model of the *A-Mab* case study (CMC Biotech Working Group, 2009). Novacyte Biologics and all sites, SOP/AMV numbers and signatories are fictional. For NLP corpus development only; not for any regulated use.

Approvals

Table 1: Approval record (synthetic; signatures not applied).

Role	Function	Name (synthetic)	Signature	Date
Author	Process Development / MSAT	—	—	—
Reviewer	Analytical Development	—	—	—
Reviewer	Manufacturing Science	—	—	—
Reviewer	Statistics	—	—	—
Approver	Quality Assurance	—	—	—

Abbreviations

AEX, anion exchange chromatography; BLA, Biologics License Application; CEX, cation exchange chromatography; CPP, critical process parameter; CQA, critical quality attribute; DNA, deoxyribonucleic acid; DoE, design of experiments; GPP, general process parameter; HCP, host cell protein; HMW, high molecular weight; ICH, International Council for Harmonisation; MSAT, Manufacturing Science and Technology; MVM, minute virus of mice; NOR, normal operating range; PAR, proven acceptable range; RPN, risk priority number; SEC, size exclusion chromatography; SOP, standard operating procedure; UF/DF, ultrafiltration and diafiltration; WC-CPP, well-controlled critical process parameter; XMuLV, xenotropic murine leukaemia virus.

1 Purpose and scope

This document records the risk assessment made on the A-Mab drug substance process before any characterization study was executed. Its output is the scope of the characterization campaign. For every process parameter in the transferred process description it names the quality attributes the parameter could affect, the potential failure mode and its effect, an initial risk score, and the type of study assigned to the parameter. The assessment is prospective throughout and reports no result.

The assessment does not classify parameters. A parameter is classified as a CPP, a WC-CPP, a key process parameter or a GPP from the effects the characterization studies measure, under SOP-4001, and the classification is reported in the per unit operation reports and consolidated in PCMR-001. The scores given below are initial scores in the same sense. The risk that remains once the studies have reported is a separate assessment made against data that does not yet exist, and nothing in this document anticipates it.

The scope is the drug substance process from the production bioreactor to formulation. Table 1.1 lists the 8 unit operations and the role each has in the control strategy. In total 37 process parameters are carried into the assessment across those unit operations. The seed expansion train, drug product manufacture and the analytical methods themselves lie outside the scope of this document.

Table 1.1: The A-Mab drug substance process train and the role of each unit operation in the control strategy.

Step	Unit operation	Principal role
3	Production Bioreactor	Forms the glycan, charge-variant and aggregate CQAs (design-space step)
4	Harvest and Clarification	Primary recovery and clarification; forms no product-quality CQA
5	Protein A Chromatography	Capture; sets leached Protein A; principal HCP and DNA clearance
6	Low-pH Viral Inactivation	Low-pH hold; sets the cumulative XMuLV (enveloped) clearance
7	Cation Exchange Chromatography	Polish; principal aggregate reduction; major HCP/DNA/leached-PA clearance
8	Anion Exchange Chromatography	Flow-through polish; sets the cumulative MVM clearance; clears XMuLV/HCP/DNA
9	Small-Virus Retentive Filtration	Dedicated small-virus removal; principal MVM clearance

Step	Unit operation	Principal role
10	Ultrafiltration / Diafiltration	Formulation / mass balance; delivers drug substance at target concentration

PTP-001 is the parent of this assessment and defines the transfer it supports. Table 1.2 lists the documents this assessment sits between. Its consumers are the characterization plans, one for each unit operation, which take the study type assigned here and set out how that study will be run. §7 names the consumer of each part of the output.

Table 1.2: Related documents.

Document ID	Title	Relationship
PTP-001	Process Transfer Plan (A-Mab Drug Substance)	Parent — transfer scope
PCMP-001	Process Characterization Master Plan (A-Mab Drug Substance)	Parent — master plan
PCMR-001	Process Characterization Master Report (A-Mab Drug Substance)	Rolls up into

Risk assessment in this framework is a lifecycle activity and not a single exercise (International Council for Harmonisation 2023b; CMC Biotech Working Group 2009). This assessment is the first in that lifecycle and draws on prior knowledge alone, since no A-Mab characterization data exist at the time it is made. It is revisited when the characterization reports are available, and again once commercial scale experience has been gained. The approach follows the enhanced development approach of ICH Q8 and ICH Q11 and supports Stage 1 of the process validation lifecycle (U.S. Food and Drug Administration 2011).

2 Risk assessment methodology

The method is risk ranking and filtering, applied to process parameters as in the A-Mab case study (CMC Biotech Working Group 2009). Each parameter is put to two questions. The first asks what the parameter could do to a quality attribute if it moved outside the range over which the process is currently understood. The second asks whether the parameter could act together with another parameter of the same unit operation, so that the size of its effect depends on where that other parameter sits. The first question sets the score. The second sets the study type, because an effect that depends on a second parameter cannot be found by moving one parameter at a time.

Severity is taken from the attribute at risk and not from the parameter. It runs from 2 to 10 and is the severity assigned to the attribute in the quality attribute assessment, reproduced in Table 3.1. Where a parameter could affect more than one attribute, the severity of the most severe of them is used. A parameter that acts only on process performance takes a severity of 4, the value assigned to a process attribute.

Occurrence and detection are assigned before any effect has been measured, and they are assigned by class rather than parameter by parameter. Occurrence is scored at 7 for a parameter that could affect a quality attribute, because neither the size of its effect nor the width of the range that holds the attribute in limit is known at this point. It is scored at 4 for a parameter that acts only on process performance, where the platform history bounds how often the parameter leaves its range. Detection is scored on what the existing controls would reveal. A failure that acts on viral clearance is scored at 10, since no in-process control and no release test on the drug substance measures the log reduction a step achieves. A failure that acts on another quality attribute is scored at 8, since release testing measures the attribute but monitoring during the unit operation does not. A failure that acts only on process performance is scored at 6, since it is seen in process or at the next unit operation.

The initial risk priority number is the product of the three scores. Across the register it runs from 96 to 700 and takes 5 distinct values, because occurrence and detection are assigned by class. Severity is therefore what separates one parameter from another within a unit operation. The number ranks parameters and sets priority, and no parameter is dispositioned on the number alone. Priority is High where an attribute of severity 8 or above is at risk, Medium where a quality attribute of lower severity is at risk, and Low where only process performance is at risk.

The assessment assigns 3 study types. A parameter carried into the designed experiment of its unit operation is studied multivariately, which is the only way to estimate an effect that depends on a second parameter. A parameter that could affect a quality attribute but is not carried into the design is studied one at a time, and the reason it was left out of the design is recorded with it. A parameter that acts only on process performance is studied one at a time. The assignment is the decision this document makes, and each characterization plan states how the assigned study will be executed.

Criticality is treated as a continuum in this framework and not as a binary split (International Council for Harmonisation 2023b; CMC Biotech Working Group 2009). The ranking in Table 4.1 therefore orders parameters against each other. It does not divide them into a critical and a non-critical group, and it is not a control strategy. Both of those follow from the studies this assessment scopes.

3 Quality attributes at risk

The severity axis of the assessment comes from the quality attribute register. Table 3.1 gives the 10 drug substance attributes the process must control, their acceptance criteria, the criticality assigned to each and the unit operation that sets each one. Criticality is the product of impact and uncertainty, which is Tool #1 of the A-Mab attribute assessment (CMC Biotech Working Group 2009). Attributes of high and very high criticality are called critical, and 6 of the 10 attributes are critical on that definition.

Table 3.1: Drug substance quality attributes, their criticality and the severity carried into this assessment.

Quality attribute	Category	Acceptance	Criticality	Tool #1	Severity	Set by
Afucosylation	glycosylation	2–13 % of glycans	H	60	8	Production Bioreactor
Galactosylation (total %Gal)	glycosylation	10–40 % (G1+G2 equiv.)	H	48	8	Production Bioreactor
High Mannose	glycosylation	3–10 % of glycans	VH	80	10	Production Bioreactor
Aggregates (HMW)	size variant	0–5 % HMW (SEC)	H	60	8	Production Bioreactor
Acidic charge variants (deamidation)	charge variant	18–40 % (CEX/icIEF)	VL	4	4	Production Bioreactor
Host Cell Protein (HCP)	process impurity	0–100 ng/mg	M-H	36	6	Production Bioreactor
Residual DNA	process impurity	0–0.001 ng/dose	VL	6	2	Production Bioreactor
Leached Protein A	process impurity	0–5 ppm	L-M	16	4	Protein A Chromatography
Viral clearance — XMuLV (enveloped)	viral safety	16.7–30 log10 (cumulative)	VH	20	10	Low-pH Viral Inactivation
Viral clearance — MVM (parvovirus)	viral safety	8.6–20 log10 (cumulative)	VH	20	10	Anion Exchange Chromatography

Of the 10 attributes, 7 are formed in the production bioreactor. The remaining 3 are set downstream, one at the capture step and one at each of the steps that fix a cumulative viral clearance requirement. That distribution is the reason the bioreactor carries more parameters into multivariate study than any other unit operation, and it is visible again in the ranking of §4.

The top severity of 10 is carried by 3 attributes. High mannose carries the highest Tool #1 score in the register at 80, because it acts on clearance and on effector function and the relationship carries real uncertainty. Both viral clearance attributes carry the same severity for a different reason. Their requirement is cumulative across the low pH hold, the anion exchange step and the virus filter, and a shortfall at any one of them is not measurable on the drug substance (International Council for Harmonisation 2023a). Severity for these two attributes is therefore assigned on the consequence of a shortfall and not on the probability of detecting one.

Acidic charge variants sit at the other end of the register with a Tool #1 score of 4 and a severity of 4. They are still carried into the assessment, because several bioreactor parameters act on them and the range over which they stay in limit is one of the things the bioreactor study must establish. Residual DNA is treated the same way. A low severity changes the priority of a parameter and does not remove it from study.

4 Parameter risk ranking by unit operation

Table 4.1 gives the ranking for the whole train. Of the 37 parameters, 21 are scored against at least one quality attribute and 16 against process performance alone. The ranking puts 16 parameters at High priority, 5 at Medium and 16 at Low. The subsections that follow take each unit operation in turn and give the potential failure mode and effect behind each score, together with the interaction judgement that decides the study type.

Table 4.1: Pre-characterization parameter risk ranking, by unit operation. Scores are initial scores and no residual risk is reported.

Step	Unit operation	Parameter	Sev.	Init. RPN	Study type	Priority
3	Production Bioreactor	Culture pH	10	560	multivariate DoE	High
3	Production Bioreactor	Culture temperature	8	448	multivariate DoE	High
3	Production Bioreactor	Dissolved CO2 (pCO2)	8	448	multivariate DoE	High
3	Production Bioreactor	Osmolality	8	448	multivariate DoE	High
3	Production Bioreactor	Culture duration	8	448	multivariate DoE	High
3	Production Bioreactor	Dissolved oxygen	4	96	univariate	Low
3	Production Bioreactor	Initial viable cell conc.	4	96	univariate	Low
3	Production Bioreactor	Basal medium concentration	4	96	univariate	Low
3	Production Bioreactor	Nutrient feed-1 volume	4	96	univariate	Low
4	Harvest and Clarification	Centrifugation (rcf)	4	96	univariate	Low
4	Harvest and Clarification	Depth filter load	4	96	univariate	Low
4	Harvest and Clarification	Post-clarification turbidity	4	96	univariate	Low
5	Protein A Chromatography	Protein load	6	336	multivariate DoE	Medium
5	Protein A Chromatography	Elution buffer pH	6	336	multivariate DoE	Medium

Step	Unit operation	Parameter	Sev.	Init. RPN	Study type	Prior-ity
5	Protein A Chromatography	Load flow rate	4	96	multivariate DoE	Low
5	Protein A Chromatography	End of pool collect	4	96	multivariate DoE	Low
5	Protein A Chromatography	Operating temperature	4	96	univariate	Low
5	Protein A Chromatography	Bed height	4	96	univariate	Low
6	Low-pH Viral Inactivation	Inactivation pH	10	700	multivariate DoE	High
6	Low-pH Viral Inactivation	Hold time	10	700	multivariate DoE	High
6	Low-pH Viral Inactivation	Temperature	10	700	multivariate DoE	High
6	Low-pH Viral Inactivation	A-Mab concentration	4	96	univariate	Low
7	Cation Exchange Chromatography	Protein load	8	448	multivariate DoE	High
7	Cation Exchange Chromatography	Load/Wash conductivity	6	336	multivariate DoE	Medium
7	Cation Exchange Chromatography	Elution buffer pH	8	448	multivariate DoE	High
7	Cation Exchange Chromatography	Elution stop collect	8	448	multivariate DoE	High
7	Cation Exchange Chromatography	Elution flow rate	4	96	univariate	Low
8	Anion Exchange Chromatography	Load pH	10	700	multivariate DoE	High
8	Anion Exchange Chromatography	Equil/Wash-1 conductivity	6	336	multivariate DoE	Medium
8	Anion Exchange Chromatography	Load conductivity	10	700	multivariate DoE	High
8	Anion Exchange Chromatography	Protein load	6	336	multivariate DoE	Medium
8	Anion Exchange Chromatography	Operating flow rate	10	700	univariate (justified)	High
9	Small-Virus Retentive Filtration	Filtration volume (load)	10	700	multivariate DoE	High
9	Small-Virus Retentive Filtration	Filtration pressure	10	700	multivariate DoE	High
10	Ultrafiltration / Diafiltration	Number of diavolumes	4	96	univariate	Low

Step	Unit operation	Parameter	Sev.	Init. RPN	Study type	Prior- ity
10	Ultrafiltration / Diafiltration	Transmembrane pressure	4	96	univariate	Low
10	Ultrafiltration / Diafiltration	Final DS concentration	4	96	univariate	Low

4.1 Production Bioreactor

The production bioreactor forms the glycan, charge variant and aggregate attributes of the drug substance, so it carries the highest severity of any step in the train. Of its 9 parameters, 5 are scored against a quality attribute, and the highest initial risk priority number in the step is 560.

Culture pH and culture duration are each scored against 5 attributes, more than any other parameter in the register. Culture pH takes a severity of 10 because high mannose is one of the attributes it puts at risk. A pH excursion shifts glycosylation, and afucosylation and galactosylation are the attributes through which A-Mab exerts its effector function, so the consequence of a shift reaches efficacy and not only conformance. Culture duration acts through culture viability. A culture extended beyond its range, or harvested once viability has fallen, lowers afucosylation and galactosylation and raises host cell protein, DNA and aggregate as lysis releases intracellular material.

Culture temperature, Dissolved CO2 (pCO2) and osmolality are each scored against galactosylation and acidic charge variants, and temperature is scored against afucosylation as well. Temperature is expected to act together with pH on the glycosylation attributes, since both act on the same glycosyltransferase activity in the culture. That judgement is why the 5 quality-linked parameters of this step are assessed in one designed experiment and not one at a time.

The remaining 4 parameters act on peak and integral viable cell concentration and on titer, and none is scored against a quality attribute. Basal medium concentration is the one that needed argument. Platform experience shows a statistically significant but shallow effect on afucosylation, too small to move the attribute across its acceptance range over the concentrations the process uses. The parameter is therefore carried on process performance grounds and assessed one at a time, and the bioreactor study reports afucosylation across the range anyway.

Table 4.2: Production bioreactor: potential failure modes and their effects.

Parameter	Potential failure mode	Potential effect	CQA(s) at risk
Culture pH	pH excursion outside the characterized range (6.6-7.1)	Shifts glycosylation (ADCC via afucosylation; CDC via galactosylation) and charge/aggregate profile	Afucosylation, Galactosylation (total %Gal), High Mannose, Acidic charge variants (deamidation), Aggregates (HMW)
Culture temperature	Temperature excursion outside 34-36 C	Alters afucosylation, galactosylation and acidic charge variants; interacts with pH	Afucosylation, Galactosylation (total %Gal), Acidic charge variants (deamidation)
Dissolved CO2 (pCO2)	Dissolved CO2 outside 40-160 mmHg	Strong effect on galactosylation and acidic variants	Galactosylation (total %Gal), Acidic charge variants (deamidation)
Osmolality	Osmolality outside 360-440 mOsm	Affects galactosylation and acidic variants	Galactosylation (total %Gal), Acidic charge variants (deamidation)
Culture duration	Culture extended beyond 15-19 days / harvested on low viability	Lowers afucosylation & galactosylation; raises HCP/DNA and aggregate as viability declines	Afucosylation, Galactosylation (total %Gal), Aggregates (HMW), Host Cell Protein (HCP), Residual DNA
Dissolved oxygen	DO outside 30-70%	Impacts peak VCD, titer and process consistency (no significant CQA impact)	process performance
Initial viable cell conc.	Inoculation density outside 0.5-1.5e6/mL	Drives peak/integral VCD and titer; no product-quality impact	process performance
Basal medium concentration	Basal medium concentration outside 0.8-1.6X	Statistically significant but shallow effect on afucosylation; drives titer	process performance
Nutrient feed-1 volume	Nutrient feed-1 volume outside 9-15% WV	Slight effect on galactosylation; critical for titer	process performance

4.2 Harvest and Clarification

Harvest recovers and clarifies the culture and forms no product quality attribute. All 3 of its parameters are scored against process performance alone and all sit at the floor of the register at 96. No parameter of this step is carried into a designed experiment.

The failure modes are a centrifugation speed outside its range, a depth filter loaded beyond its capacity, and a high turbidity after clarification. Each acts on clarification efficiency, on filter capacity or on the quality of the feed presented to the capture step, and each is visible in the turbidity in-process control before the material moves on. The capture step is robust to the feedstock over the range this step produces, which is what keeps a harvest failure a performance failure rather than a quality failure.

Table 4.3: Harvest and clarification: potential failure modes and their effects.

Parameter	Potential failure mode	Potential effect	CQA(s) at risk
Centrifugation (rcf)	Centrifugation rcf outside 6000-12000 g	Affects clarification/yield (no CQA impact)	process performance
Depth filter load	Depth filter load outside 80-160 L/m2	Affects filter capacity/turbidity (no CQA impact)	process performance
Post-clarification turbidity	Post-clarification turbidity high	Feed quality to Protein A (no CQA impact)	process performance

4.3 Protein A Chromatography

Capture binds the antibody from the clarified harvest and removes the bulk of the host cell protein and DNA. Protein load and elution buffer pH are the 2 parameters scored against a quality attribute, both against host cell protein at a severity of 6. A load above the characterized range raises the host cell protein level of the pool, and how far it rises depends on where the elution pH sits. Both are therefore assessed together. Elution buffer pH also sets the feed to the low pH hold, which is the one feed forward link in this part of the train and is stated again in §4.4.

Load flow rate and end of pool collect carry no quality attribute and act on step yield. Both are nevertheless carried into the capture design, because the same design bounds yield and they cost no additional runs once the design exists. Operating temperature and bed height are assessed one at a time.

No operating parameter of this step is scored against leached Protein A, although the attribute is set here. Leached Protein A is governed by the resin and by its life cycle, packing and sanitization under SOP-2008, and not by a parameter varied within its operating range, and the polishing steps clear what reaches them. The attribute is therefore held by material and procedural controls. The capture study measures it in the pool across the designed experiment, so the ranges this assessment scopes are bounded against it even though no parameter is ranked on it.

Table 4.4: Protein A chromatography: potential failure modes and their effects.

Parameter	Potential failure mode	Potential effect	CQA(s) at risk
Protein load	Protein load above characterized range (>50 g/L resin)	Increases pool HCP (interaction with elution pH)	Host Cell Protein (HCP)
Elution buffer pH	Elution pH below characterized range (<3.2)	Increases pool HCP; also sets up viral inactivation feed	Host Cell Protein (HCP)
Load flow rate	Load flow rate high with high load	Reduces product yield (no CQA impact)	process performance
End of pool collect	End-of-pool collection outside 2.0-3.2 CV	Affects product yield (no CQA impact)	process performance
Operating temperature	Parameter outside characterized range	No significant product-quality impact expected	process performance
Bed height	Parameter outside characterized range	No significant product-quality impact expected	process performance

4.4 Low-pH Viral Inactivation

The low pH hold sets the cumulative XMuLV clearance of the process. Of its 4 parameters, 3 reach the top of the register at 700, which is the highest initial risk priority number in the assessment.

Inactivation pH carries a failure mode in both directions. Above the range, inactivation of enveloped virus is incomplete. Below it, the antibody is exposed to conditions under which aggregation and precipitation have been seen on the platform. Hold time carries the same risk in both directions, since a short hold leaves inactivation incomplete and a long hold raises aggregate. Temperature acts on the rate of inactivation, so a temperature below the range matters most at the short end of the hold and the high end of the pH range. The parameters act on the same outcome through the same kinetics, which is the reason they are assessed in one designed experiment.

Detection is at the top of the scale for each of them. Neither an in-process control nor a release test on the drug substance measures the log reduction the hold achieves, and the claim is made instead from a spiking study on a qualified small scale model (International Council for Harmonisation 2023a). That is what carries these parameters above the bioreactor parameters in the ranking, even though the bioreactor attributes carry the same severity.

A-Mab concentration is scored against process performance. Prior product experience shows no effect of concentration on inactivation, on aggregation or on charge variants over the range the pool reaches, so the parameter is assessed one at a time.

Table 4.5: Low-pH viral inactivation: potential failure modes and their effects.

Parameter	Potential failure mode	Potential effect	CQA(s) at risk
Inactivation pH	pH > 4.0 (incomplete inactivation) or pH < 3.2 (aggregation/precipitation)	Loss of XMuLV inactivation assurance or increased aggregate	Viral clearance — XMuLV (enveloped), Aggregates (HMW)
Hold time	Hold time < 60 min (incomplete) or > 180 min (aggregation)	Incomplete inactivation or increased aggregate	Aggregates (HMW), Viral clearance — XMuLV (enveloped)
Temperature	Temperature < 15 C	Lower inactivation rate at short time / higher pH	Viral clearance — XMuLV (enveloped)
A-Mab concentration	Concentration > 35 g/L	No significant effect on inactivation, aggregation or charge variants	process performance

4.5 Cation Exchange Chromatography

The cation exchange step is the principal aggregate reduction step of the process and also clears host cell protein, DNA and leached Protein A. It sets no attribute of its own. Of its 5 parameters, 4 are scored against a quality attribute, and protein load carries the highest severity of them at 8 because both aggregate and host cell protein are at risk.

Aggregate is the attribute that matters most here. No step after the cation exchange step is credited with aggregate reduction, so aggregate that leaves this step in the pool reaches the drug substance. Protein load, elution buffer pH and elution stop collect all act on the aggregate content of the pool, and load and elution pH act on it together, since the resolution between monomer and aggregate in the elution peak depends on both. Load and wash conductivity is scored against host cell protein, where a conductivity below the range weakens the wash and leaves more host cell protein bound to be eluted with the product.

Elution flow rate is scored against process performance. It acts on peak shape and on yield, and any effect on aggregate is a second order consequence of peak shape, so the parameter is assessed one at a time.

Table 4.6: Cation exchange chromatography: potential failure modes and their effects.

Parameter	Potential failure mode	Potential effect	CQA(s) at risk
Protein load	Protein load above characterized range (>40 g/L resin)	Reduced aggregate and HCP clearance (interactions with pH and wash conductivity)	Aggregates (HMW), Host Cell Protein (HCP)
Load/Wash conductivity	Load/wash conductivity below 3 mS/cm	Reduced HCP clearance	Host Cell Protein (HCP)
Elution buffer pH	Elution pH outside 5.8-6.2	Alters aggregate distribution and yield	Aggregates (HMW)
Elution stop collect	Extended elution stop-collect (>1.5 OD)	Minor increase in pool aggregate	Aggregates (HMW)
Elution flow rate	Elution flow rate outside 100-300 cm/hr	Affects peak shape and yield (minor aggregate)	process performance

4.6 Anion Exchange Chromatography

The anion exchange step is operated in flow-through mode. Impurities and virus bind to the resin while the antibody passes through, and the step sets the cumulative MVM clearance of the process. All 5 of its parameters are scored against a quality attribute, which is true of no other unit operation in the train.

Load pH is scored against 3 attributes and reaches 700. It governs the charge on both the impurities and the resin, so a load pH below the characterized range weakens the binding of host cell protein and of both model viruses at once. Load conductivity acts on the same binding through ionic strength and is scored against both viral attributes, and it acts together with load pH, since the two set the binding condition jointly. Equil and wash-1 conductivity and protein load are scored against host cell protein, where the risk is a loss of capacity rather than a loss of clearance.

Operating flow rate is the one parameter in the register with a quality attribute at risk that is not carried into a designed experiment. It acts on the step only through residence time and is not expected to act together with the load chemistry, so a study that moves it one at a time bounds it. The parameter is assigned a justified univariate study on that reasoning, and PCP-008 records the justification with the design. That is a judgement made from platform experience and not a measurement, and §6 states what follows if it is wrong.

Table 4.7: Anion exchange chromatography: potential failure modes and their effects.

Parameter	Potential failure mode	Potential effect	CQA(s) at risk
Load pH	Load pH below characterized range (<7.2)	Reduced HCP removal and reduced XMuLV/MVM clearance	Host Cell Protein (HCP), Viral clearance — XMuLV (enveloped), Viral clearance — MVM (parvovirus)
Equil/Wash-1 conductivity	Equil/Wash-1 conductivity above 3.6 mS/cm	Reduced HCP removal	Host Cell Protein (HCP)
Load conductivity	Load conductivity above characterized range	Reduced viral clearance (interacts with pH)	Viral clearance — XMuLV (enveloped), Viral clearance — MVM (parvovirus)
Protein load	Protein load above characterized range (>300 g/L resin)	Reduced HCP removal capacity (viral clearance robust)	Host Cell Protein (HCP)
Operating flow rate	Operating flow rate above 450 cm/hr	Reduced residence time / viral clearance margin	Viral clearance — XMuLV (enveloped), Viral clearance — MVM (parvovirus)

4.7 Small-Virus Retentive Filtration

The virus filter removes small non-enveloped virus by size and is the principal MVM clearance step. Both of its parameters are scored against MVM clearance and both reach 700. The step carries no parameter scored against process performance alone, which is unusual in this train and follows from the step having one purpose.

Volumetric load is the parameter the step turns on. Log reduction falls as the volume filtered per unit area rises, because the retentive layer accumulates material and the probability of parvovirus passage rises with it. Filtration pressure is scored against both model viruses. Pressure outside the filter operating limits may act on virus removal, and the risk is largest at high load, where the two parameters act together. Both are therefore carried into the designed experiment.

Table 4.8: Small-virus retentive filtration: potential failure modes and their effects.

Parameter	Potential failure mode	Potential effect	CQA(s) at risk
Filtration volume (load)	Volumetric load above 105 L/m2	Reduced MVM log-reduction (parvovirus breakthrough risk)	Viral clearance — MVM (parvovirus)
Filtration pressure	Pressure outside filter operating limits	Potential impact on virus removal at high load	Viral clearance — MVM (parvovirus), Viral clearance — XMuLV (enveloped)

4.8 Ultrafiltration / Diafiltration

The formulation step concentrates the antibody and exchanges it into the drug substance buffer. It forms and clears no quality attribute. All 3 of its parameters are scored against process performance and sit at the floor of the register at 96.

A diavolume count below the range leaves the buffer exchange incomplete. A transmembrane pressure outside its range acts on flux and on processing time. The final concentration is the drug substance concentration itself, which is measured directly and released against a specification, so a failure of this parameter is detected before the batch is dispositioned. Each is assessed one at a time.

Table 4.9: Ultrafiltration and diafiltration: potential failure modes and their effects.

Parameter	Potential failure mode	Potential effect	CQA(s) at risk
Number of diavolumes	Diavolumes < 7	Incomplete buffer exchange (formulation; not part of DS characterization)	process performance
Transmembrane pressure	TMP outside 10-15 psi	Affects flux/processing (formulation)	process performance
Final DS concentration	Final concentration outside 65-85 g/L	DS concentration spec (formulation)	process performance

5 Characterization study assignment

The assignment is the output of this assessment. Of the 37 parameters, 22 are assigned to a multivariate design and 15 to univariate assessment. A designed experiment is run at 6 of the 8 unit operations. The other 2 are harvest and formulation, neither of which is scored against a quality attribute, and no designed experiment is to be run for them. Table 5.1 gives the split per unit operation with the plan and report that cover each.

Table 5.1: Characterization scope per unit operation, and the documents that cover it.

Step	Unit operation	Parameters	Multivariate	Univariate	Documents
3	Production Bioreactor	9	5	4	PCP-003 / PCR-003
4	Harvest / Clarification	3	0	3	PCP-004 / PCR-004
5	Protein A Chromatography	6	4	2	PCP-005 / PCR-005
6	Low-pH Viral Inactivation	4	3	1	PCP-006 / PCR-006
7	Cation Exchange Chromatography	5	4	1	PCP-007 / PCR-007
8	Anion Exchange Chromatography	5	4	1	PCP-008 / PCR-008
9	Small Virus Retentive Filtration	2	2	0	PCP-009 / PCR-009
10	Ultrafiltration / Diafiltration (formulation)	3	0	3	PCP-010 / PCR-010

The parameters assigned to a multivariate design are those judged able to act together with another parameter of the same unit operation. The judgement was made from platform and prior product experience and it is recorded per parameter in §4. The capture step carries 2 parameters into a design although they are scored against process performance alone, because the same design bounds step yield and their inclusion costs no additional runs.

The assessment assigns 1 parameter to a justified univariate study. It is the operating flow rate of the anion exchange step, which is scored against both viral clearance attributes and acts on the step only through residence time. The justification is that a parameter which does not act together with the load chemistry can be bounded one at a time, and PCP-008 carries it with the design.

No parameter is excluded from study. The lowest ranking parameters act on process performance rather than on a quality attribute, and they are studied one at a time so that the ranges entered in the batch record are supported by data rather than by platform precedent alone. A risk assessment of this kind may filter parameters out of study altogether, and that option was considered and not taken here. The parameter list is small enough that the cost of assessing the low ranking parameters is a univariate study each.

Table 5.2 gives the parameter set handed to the plans, with the set-point, the range to be studied and the normal operating range. For 31 of the 37 parameters the range to be studied is wider than the normal operating range on both sides, so that the edges of the characterized region lie outside the region in which the process will run (CMC Biotech Working Group 2009). For the remaining 6 the two ranges share a bound, because that bound is a limit of the step rather than a control range. A minimum hold time and a maximum volumetric load are the clearest cases, and neither can be exceeded in the direction that would widen the study.

Table 5.2: The parameter set handed to the characterization plans, with the range to be studied.

Step	Parameter	Unit	Set-point	Range studied	NOR	Study type
3	Culture pH	pH	6.85	6.6–7.1	6.75–6.95	multivariate
3	Culture temperature	°C	35	34–36	34.5–35.5	multivariate
3	Dissolved CO2 (pCO2)	mmHg	100	40–160	70–130	multivariate
3	Osmolality	mOsm	400	360–440	375–425	multivariate
3	Culture duration	days	17	15–19	16–18	multivariate
3	Dissolved oxygen	% sat	50	30–70	40–60	univariate
3	Initial viable cell conc.	e6 cells/mL	1	0.5–1.5	0.8–1.2	univariate
3	Basal medium concentration	X	1.2	0.8–1.6	1–1.4	univariate
3	Nutrient feed-1 volume	% WV	12	9–15	10–14	univariate
4	Centrifugation (rcf)	g	9000	6000–12000	8000–10000	univariate
4	Depth filter load	L/m2	120	80–160	100–140	univariate
4	Post-clarification turbidity	NTU	5	1–20	1–10	univariate
5	Protein load	g/L resin	30	10–50	20–40	multivariate

Step	Parameter	Unit	Set-point	Range studied	NOR	Study type
5	Elution buffer pH	pH	3.55	3.2–3.9	3.4–3.7	multivariate
5	Load flow rate	cm/hr	200	100–300	150–250	multivariate
5	End of pool collect	CV	2.6	2–3.2	2.3–2.9	multivariate
5	Operating temperature	°C	22	15–30	18–26	univariate
5	Bed height	cm	20	10–30	15–25	univariate
6	Inactivation pH	pH	3.5	3.2–4	3.4–3.6	multivariate
6	Hold time	min	90	60–180	60–120	multivariate
6	Temperature	°C	21	15–25	19–23	multivariate
6	A-Mab concentration	g/L	20	5–35	5–35	univariate
7	Protein load	g/L resin	25	10–40	18–32	multivariate
7	Load/Wash conductivity	mS/cm	5	3–7	4–6	multivariate
7	Elution buffer pH	pH	6	5.8–6.2	5.85–6.15	multivariate
7	Elution stop collect	OD desc.	1	0.5–1.5	0.7–1.3	multivariate
7	Elution flow rate	cm/hr	200	100–300	150–250	univariate
8	Load pH	pH	7.5	7.2–7.8	7.35–7.65	multivariate
8	Equil/Wash-1 conductivity	mS/cm	2.6	1.6–3.6	2.1–3.1	multivariate
8	Load conductivity	mS/cm	5.5	3–8	4–7	multivariate
8	Protein load	g/L resin	200	50–300	120–280	multivariate
8	Operating flow rate	cm/hr	300	150–450	200–400	univariate
9	Filtration volume (load)	L/m2	90	50–140	50–105	multivariate
9	Filtration pressure	psi	13	8–30	10–20	multivariate
10	Number of diavolumes	DV	8	7–10	7–9	univariate

Step	Parameter	Unit	Set-point	Range studied	NOR	Study type
10	Transmembrane pressure	psi	12	10–15	10–14	univariate
10	Final DS concentration	g/L	75	65–85	70–80	univariate

The assignment is prospective. It is revisited when the characterization reports are available, and a parameter whose measured effect contradicts the judgement made here is reported as such in the report of its unit operation.

6 Assumptions and limitations

The assessment assumes that platform behaviour transfers to A-Mab. The mechanisms it relies on were established on the prior humanized IgG1 antibodies X-Mab, Y-Mab and Z-Mab and on A-Mab clinical and engineering experience, and the assumption is that A-Mab responds to the same parameters in the same directions (CMC Biotech Working Group 2009). Where it does not, the characterization study finds it, which is one of the reasons the studied ranges are wider than the operating ranges.

The parameter list is assumed complete. It is the list carried in the process description transferred under PTP-001, and a parameter that is not in that description is not assessed here. An equipment or facility difference at the receiving site that introduces a new parameter is handled under the transfer plan and would require this assessment to be revised.

Severity is taken from the quality attribute assessment and is not re-derived. That assessment scores impact and uncertainty and deliberately excludes manufacturing capability and detectability (CMC Biotech Working Group 2009), so a severity in Table 3.1 says what a failure would mean for the patient and says nothing about how likely the process is to produce one.

Occurrence and detection are assigned by class rather than parameter by parameter, so the initial risk priority number takes only 5 values across 37 parameters. The number therefore separates classes of parameter well and separates parameters within a class not at all. Ranking within a unit operation should be read from severity and from the reasoning in §4.

The interaction judgement is the weakest part of the assessment. A designed experiment estimates interactions only among the parameters it carries, so an interaction between a parameter in the design and one left out of it cannot be found by that design. The clearest exposure is the operating flow rate of the anion exchange step, which is scored against both viral clearance attributes and studied one at a time. If the anion exchange study finds that residence time acts together with the load chemistry, the univariate result for flow rate would not bound the step and the design would have to be augmented.

Interactions across unit operations are not scored. Each parameter is assessed against the attributes of its own step. The feed forward links in the train are real, and the elution pH of the capture step setting the feed to the low pH hold is the clearest of them, but they are handled by the in-process limits of the receiving step rather than by a score in Table 4.1.

Finally, no number in this document is a measurement of A-Mab process behaviour. The scores are judgements, the studies are not yet run, and the ranges in Table 5.2 are the ranges to be studied and not ranges that have been shown to be acceptable.

7 Outputs and downstream use

This assessment produces three things. The ranking in Table 4.1 is the pre-characterization risk basis. The study type in Table 4.1 and the summary in Table 5.1 are the scope. The parameter set in Table 5.2 is what each plan instantiates.

Per unit operation the consumer is the characterization plan named in Table 5.1. Each plan cites this assessment as its risk basis, takes the study type and the range to be studied, and adds the scale-down model, the analytical methods, the sampling plan and the acceptance criteria. The report of that unit operation then measures the effects, defines the operating region and classifies each parameter under SOP-4001.

PCMP-001 consumes the assessment at campaign level. It states once what the individual plans would otherwise each restate, including the count of parameters going to multivariate and to univariate study across the train. PCMR-001 consolidates the reports and is where the risk that remains after characterization is addressed.

This document is superseded by its own revision. A change to the parameter list, to a severity, or to an assigned study type is made by revising this assessment, and the affected plans are revised with it.

8 Approvals

This assessment is approved by the functions listed in Table 1. Approval confirms that the parameter list is complete against the transferred process description, that the severity carried for each attribute agrees with the quality attribute assessment, and that the study type assigned to each parameter is supported by the reasoning given in §4. A change to the parameter list or to an assigned study type after approval is made by revision of this document.

References

CMC Biotech Working Group. 2009. *A-Mab: A Case Study in Bioprocess Development*. CASSS.

International Council for Harmonisation. 2023a. *ICH Q5A(R2): Viral Safety Evaluation of Biotechnology Products Derived from Cell Lines of Human or Animal Origin*. ICH.

International Council for Harmonisation. 2023b. *ICH Q9(R1): Quality Risk Management*. ICH.

U.S. Food and Drug Administration. 2011. *Guidance for Industry: Process Validation — General Principles and Practices*. FDA.